

Homework 4

MATH 346/377 - Winter 2025

Due on Sunday, March 16, 9:00pm on Crowdmark

Math 346: Do problems 1 to 4 – Total of points= 40 pts

Math 377: Do problems 1 to 6 – Total of points= 60 pts

1. (8 points) Find all primitive roots modulo 17.

Solution: Since $\phi(17) = 16$, a primitive root modulo 17 has order 16. More precisely, g is a primitive root if and only if g is coprime to 17 and $g^{16/2} \equiv g^8 \not\equiv 1 \pmod{17}$, by a theorem that we have seen in class. Hence, we see that 2 is not a primitive root since

$$2^8 \equiv (2^4)^2 \equiv (16)^2 \equiv (-1)^2 \equiv 1 \pmod{17}.$$

On the other hand, for $g = 3$ we have

$$3^4 \equiv 81 \equiv 30 \equiv -4 \pmod{17}$$

and so $3^8 \equiv 16 \equiv -1 \pmod{17}$. It follows that 3 is a primitive root, and the $\phi(\phi(17)) = 8$ primitive roots are of the form 3^i with i coprime to 16:

$$\{3, 3^3, 3^5, 3^7, 3^9, 3^{11}, 3^{13}, 3^{15}\} \equiv \{3, 5, 6, 7, 10, 11, 12, 14\} \pmod{17}.$$

(Observe that $3^9 \equiv (3^8) \cdot 3 \equiv -3 \pmod{17}$, $3^{11} \equiv -3^3 \pmod{17}$,... So the second half of the primitive roots is just the negative of the first half. This works for any p and relies on the fact that for any primitive root modulo p we have $g^{\frac{p-1}{2}} \equiv -1 \pmod{p}$.)

2. (12 points) Find all integers $0 \leq x \leq 92$ solving the equation

$$x^9 + 4 \equiv 0 \pmod{93}.$$

Solution: Since $93 = 31 \times 3$, by the Chinese Remainder Theorem $x^9 + 4 \equiv 0 \pmod{93}$ if and only

$$x^9 + 4 \equiv 0 \pmod{31} \text{ and } x^9 + 4 \equiv 0 \pmod{3}.$$

Modulo 3 we have that $x^3 \equiv x \pmod{3}$ for any integer x , so

$$x^9 + 4 \equiv (x^3)^3 + 4 \equiv x + 1 \pmod{3}.$$

This shows that x must be congruent to 2 modulo 3.

To solve modulo 31 we have to find a primitive root modulo 31. The residue 2 is not a primitive root modulo 31 since $2^5 \equiv 32 \equiv 1 \pmod{30}$, so the order of 2 is 5. The prime divisors of $\phi(30)$ are 2, 3, 5 so g is a primitive root modulo 31 if and only if

$$g^{\frac{30}{2}} \equiv g^{15} \not\equiv 1 \pmod{31},$$

$$g^{\frac{30}{3}} \equiv g^{10} \not\equiv 1 \pmod{31},$$

$$g^{\frac{30}{5}} \equiv g^6 \not\equiv 1 \pmod{31}.$$

For $g = 3$ we find that

$$3^6 \equiv (3^3)^2 \equiv (-4)^2 \equiv 2^4 \equiv 16 \pmod{31},$$

$$3^{10} \equiv 3^6 \cdot 3^3 \cdot 3 \equiv 16 \cdot (-4) \cdot 3 \equiv -24 \equiv 25 \pmod{31},$$

$$3^{15} \equiv (3^6)^2 \cdot 3^3 \equiv 2^8 \cdot (-4) \equiv -32 \equiv -1 \pmod{31}.$$

So $g = 3$ is a primitive root modulo 31, and any x coprime to 31 can be written $x = 3^i$ for some unique $0 \leq i \leq 30$. Also note that $-4 \equiv 27 \equiv 3^3 \pmod{31}$. Now we have to find all the integers $0 \leq i \leq 30$ such that

$$3^{9i} \equiv -4 \equiv 3^3 \pmod{31}$$

which is equivalent to

$$3^{3(3i-1)} \equiv 1 \pmod{31}.$$

Since the order of 3 modulo 31 is 30, we find that we must have $3(3i - 1) \equiv 0 \pmod{30}$, which implies $3i - 1 \equiv 0 \pmod{10}$. Since the inverse of 3 modulo 10 is 7, we find that

$$i \equiv 7 \pmod{10}.$$

So the possible values of i between 0 and 30 are $\{7, 17, 27\}$, and we find that the solutions to the equation modulo 31 are

$$x \in \{3^7, 3^{17}, 3^{24}\} \pmod{31} = \{17, 22, 23\} \pmod{31}.$$

The integers $0 \leq x \leq 92$ that are congruent to 17, 22 or 23 modulo 31 are

$$\{17, 22, 23, 48, 53, 24, 79, 84, 85\}.$$

Among these 9 solutions there are three that are congruent to 2 modulo 3:

$$\{17, 23, 53\}.$$

These are all the integers $0 \leq x \leq 92$ that solve $x^9 + 4 \equiv 0 \pmod{93}$.

3. (9 points) Let p be a prime. Show that if a has order 3 modulo p , then $1 + a$ has order 6 modulo p .

Solution: If a has order 3 modulo p then $a^3 \equiv 1 \pmod{p}$ and $a \not\equiv 1 \pmod{p}$. Hence, we have

$$a^3 - 1 \equiv (a - 1)(1 + a + a^2) \equiv 0 \pmod{p}.$$

Since p does not divide $a - 1$, it must divide $1 + a + a^2$. It follows that

$$1 + a \equiv -a^2 \pmod{p},$$

and we have to show that the order of $-a^2$ is 6. We have $(-a^2)^6 \equiv (a^3)^4 \equiv 1 \pmod{p}$ so the order of a divides 6. On the other hand, we have $(-a^2)^3 \equiv -a^6 \equiv -1 \pmod{p}$, and $(-a^2)^2 \equiv a^4 \equiv a \pmod{p}$. So $\text{ord}_p(1 + a) = \text{ord}_p(-a^2) = 6$.

4. (11 points) Let p be a prime. Use primitive roots modulo p to show Wilson's theorem, that states that

$$(p - 1)! \equiv -1 \pmod{p}.$$

Solution: Let g be a primitive root. For each integer $1 \leq n \leq p - 1$ there is a unique $0 \leq i \leq p - 2$ such that

$$g^i \equiv n \pmod{p}.$$

Thus, we have

$$\begin{aligned} (p - 1)! &= 1 \cdot 2 \cdots (p - 2) \cdot (p - 1) \\ &\equiv g^0 \cdot g^1 \cdots g^{p-3} \cdot g^{p-2} \pmod{p} \\ &\equiv g^{1+2+\cdots+(p-3)+(p-2)} \pmod{p}. \end{aligned}$$

The sum of integers from 1 to $(p - 2)$ is given by $\frac{(p-1)(p-2)}{2}$, so that

$$(p - 1)! \equiv g^{\frac{(p-1)(p-2)}{2}} \equiv (g^{p-2})^{\frac{p-1}{2}} \pmod{p}.$$

Since $p - 2$ is coprime to $p - 1$, we have that $h := g^{p-2} \not\equiv 1$ is also a primitive root modulo p . Now $x = h^{\frac{p-1}{2}}$ satisfies

$$x^2 - 1 \equiv h^{p-1} - 1 \equiv 0 \pmod{p}.$$

But $x \not\equiv 1$, since h is a primitive root, so

$$-1 \equiv x \equiv h^{\frac{p-1}{2}} \equiv (p-1)! \pmod{p}.$$

Problems 5 and 6 for MATH 377 students.

5. (10 points) Let p be a prime and $k \geq 0$ an integer. Show that

$$1^k + 2^k + \cdots + (p-1)^k \equiv \begin{cases} 0 \pmod{p} & \text{if } p-1 \nmid k \\ -1 \pmod{p} & \text{if } p-1 \mid k \end{cases}$$

Solution: If $p-1$ divides k , then by FLT $n^k \equiv 1 \pmod{p}$ for all $1 \leq n \leq p-1$. Hence, the sum is

$$1^k + 2^k + \cdots + (p-1)^k \equiv 1 + 1 + \cdots + 1 \equiv p-1 \equiv -1 \pmod{p}.$$

Now assume that $p-1$ does not divide k and let us denote the sum by

$$S := \sum_{x \in (\mathbb{Z}/p\mathbb{Z})^\times} x^k.$$

Let g be a primitive root. Since multiplication by g is a bijection of $(\mathbb{Z}/p\mathbb{Z})^\times$, we have

$$g^k S \equiv \sum_{x \in (\mathbb{Z}/p\mathbb{Z})^\times} (gx)^k \equiv \sum_{y \in (\mathbb{Z}/p\mathbb{Z})^\times} y^k \equiv S \pmod{p}.$$

Thus, we find that

$$(g^k - 1)S \equiv 0 \pmod{p}.$$

If we had $g^k - 1 \equiv 0 \pmod{p}$, that would imply that $\text{ord}_p(g) = p-1$ divides k which contradicts our assumption. So $g^k - 1 \not\equiv 0 \pmod{p}$ and thus

$$S \equiv 0 \pmod{p}.$$

6. (a) (4 points) Let $n \geq 0$. Use induction to show that $5^{2^n} - 1$ is divisible by 2^{n+2} but not by 2^{n+3} .
 (b) (6 points) Deduce the order of 5 modulo 2^e for $e \geq 3$, and that

$$(\mathbb{Z}/2^e\mathbb{Z})^\times = \{\pm 5^i \mid 0 \leq i < 2^{e-2}\}$$

In particular, this exercise shows that

$$(\mathbb{Z}/2^e\mathbb{Z})^\times \simeq (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2^{e-2}\mathbb{Z})$$

is almost cyclic, whereas for an odd prime we have seen that

$$(\mathbb{Z}/p^e\mathbb{Z})^\times \simeq (\mathbb{Z}/p^{e-1}(p-1)\mathbb{Z}).$$

Solution:

- (a) When $n = 0$ then $5^1 - 1 = 4$ is divisible by $2^{0+2} = 4$ but not $2^{0+3} = 8$. Now assume that for some n we know that $5^{2^n} - 1$ is divisible by 2^{n+2} but not by 2^{n+3} . For $n + 1$ we rewrite

$$5^{2^{n+1}} - 1 = (5^{2^n})^2 - 1 = (5^{2^n} - 1)(5^{2^n} + 1).$$

The first term is divisible by 2^{n+2} but not by 2^{n+3} by the induction hypothesis. For the second term we have that

$$5^{2^n} + 1 \equiv 2 \pmod{4},$$

so it is divisible by 2 but not by 4. Together this shows that $5^{2^{n+1}} - 1$ is divisible by $2^{n+2} \cdot 2 = 2^{n+3}$ but not by $2^{n+2} \cdot 4 = 2^{n+4}$, and this proves the induction.

- (b) If we take $n = e - 2$ in part a), we find that

$$5^{2^{e-2}} - 1 \equiv 0 \pmod{2^e} \quad 5^{2^{e-2}} - 1 \not\equiv 0 \pmod{2^{e+1}}.$$

For $n = e - 3$ we find that

$$5^{2^{e-3}} - 1 \equiv 0 \pmod{2^{e-1}}, \quad 5^{2^{e-3}} - 1 \not\equiv 0 \pmod{2^e}.$$

Hence, the order of 5 modulo 2^e is 2^{e-2} . There are 2^{e-1} classes that are invertible modulo 2^e :

$$(\mathbb{Z}/2^e\mathbb{Z})^\times \simeq \{1, 3, 5, 7, \dots, 2^e - 1\}.$$

When we look at the powers of 5 we get exactly, 2^{e-2} elements

$$S_1 := \{1, 5, 5^2, \dots, 5^{2^{e-2}-1}\},$$

so half of the elements in $(\mathbb{Z}/2^e\mathbb{Z})^\times$. The other half is given by the negatives

$$S_2 := \{-1, -5, -5^2, \dots, -5^{2^{e-2}-1}\}.$$

To conclude the argument, we have to make sure that the second set is disjoint from the first one, that is to show that we can never have $-5^i \equiv 5^k \pmod{2^e}$ for some $0 \leq i, k < 2^{e-2}$. Indeed if we had $-5^i \equiv 5^k \pmod{2^e}$, then we would have

$$5^{i+j} + 1 \equiv 0 \pmod{2^e}.$$

But since $e \geq 3$ this congruence is impossible, as we have already observed that $5^{i+j} + 1 \equiv 2 \pmod{4}$.